POOJA KIRAN | SECURITY ENGINEERING

Evaluating an Offline Detector Against Adversarial Prompt Attacks

A Reproducible Framework for Prompt-Attack Generation and Defensive Evaluation

Source inventory: 2e2989a915cc013c68c46bb6995831b6344cd2a8

Test snapshot: 08b40538c00146981d450cad7af5cedd87b887b6

CI run 36783655822 | initiated 2026-09-30 UTC | MIT

https://github.com/poojakira/llm-redteam-framework

175
tests passed in cited CI

94.30%
statement coverage at cited CI

0.7188
OOD F1, 50 hand-authored texts

64%
benign false-positive rate, 16/25

CONTRIBUTION

- A reproducible offline prompt-attack evaluation framework that reports both the stronger in-distribution result and the weaker novel-phrasing result, making the detector's generalization gap explicit instead of hiding it behind a random split.

IMPLEMENTED METHOD

- 1. Generate adversarial and benign prompt corpora across defined attack categories.
- 2. Train/evaluate the offline TF-IDF + Logistic Regression detector.
- 3. Compare random, grouped-template, structural-fixture, and novel-phrasing OOD evaluations.
- 4. Emit JSON/SARIF evidence for CI.
- 5. Keep live-model behavior claims out of scope unless separately measured.

LIMITATIONS AND EVIDENCE BOUNDARY

- Offline detector evaluation does not prove whether a live LLM would follow an injected instruction.
- English/template coverage is finite.
- Structural fixtures are not equivalent to true OOD natural-language paraphrases.
- No real-world jailbreak-defense rate is claimed.

SCOPED EVIDENCE

- Cited Python 3.12 CI: 175 passed, 1 skipped; 94.30% statement coverage, same test job.
- Performance job 110120166630: OOD precision 0.5897, recall 0.92, F1 0.7188 on 25 adversarial and 25 benign hand-authored regression texts.
- Grouped-template F1 0.9714 is an easier in-distribution reference. A prior random-split F1 of 1.00 is reported in source evidence, not independently confirmed by this audit.
- Offline comparison of the same detector; these figures do not establish live LLM protection or population-level generalization.

REPRODUCTION

- git checkout 08b40538c00146981d450cad7af5cedd87b887b6
- Install the documented development dependencies; run python -m pytest tests/ -q. Use the cited job configuration when reproducing coverage.
- Benchmark data, adapters and environment requirements are separate; consult poster/13_reproducibility.md and the claim ledger.

Dated tests and committed fixtures do not establish universal detection, production deployment or an SLO.